

THESHIKA NAVOD | COMPUTER ENGINEERING

theshikanavod@gmail.com | +94713621414 | [GitHub](#) | [LinkedIn](#) | [Medium](#) | [Website](#) | Sri Lanka

PROFILE SUMMARY

Software Engineer specializing in **Java, Spring Boot, Microservices, Distributed Systems, and Cloud-native development**, with strong experience building **production-grade AI systems** including RAG pipelines, agentic architectures, and multimodal applications. Hands-on expertise in **Kafka, gRPC, CI/CD, Docker, Databases**, and scalable backend engineering. Passionate about building reliable, high-performance systems.

WORK EXPERIENCE

Applantics (PVT) LTD Associate Software Engineer February 2025 – Current

- Developing enterprise POS system with microservices architecture handling real-time transaction processing, inventory management, and payment gateway integrations for multiple retail clients using Java Spring Boot and REST APIs.
- Building AI-powered customer support chatbot using RAG architecture with LangChain and vector databases, and designing real-time analytics dashboards with React and WebSocket for business insights.

Tech Stack: Java, JavaScript, Python, React, MySQL, MongoDB, AI Agents, AI Chatbots, LangChain, Docker, AWS.

Information Systems Associates (ISA) Intern Software Engineer July 2024 – January 2025

- Developed and maintained microservices for Air Arabia's aviation systems using Spring Boot and Angular
- Refactored legacy codebase resolving 200+ critical bugs identified by SonarQube and increased unit test coverage from 10% to 80% using JUnit and Mockito, improving code maintainability by 60% and reducing production defects by 30%
- Optimized system performance by migrating REST endpoints to gRPC protocol with event streaming and implementing real-time estimate departure time display, reducing inter-service latency by 40% and passenger wait time by 25%
- Worked in Agile environment with daily standups, sprint planning, and retrospectives using Jira and Confluence

Tech Stack: Java (8, 11, 17), Spring Boot, React, Angular, PostgreSQL, Oracle DB, Redis, Kafka, gRPC, Docker, Jenkins, JUnit, Mockito, SonarQube, Git, SOAP/REST APIs, Agile environment with Jira/Confluence

PROJECT EXPERIENCE

AI Code Generator and Reviewer System - Research Project (code - upon request)

Built VS Code extension automating code generation and bug detection for legacy codebases, achieving 42% score on SWE-bench. Developed multi-agent reviewer system using fine-tuned LLMs detecting security vulnerabilities, code smells, and performance issues with human-in-the-loop validation, reducing manual review time by 60%.

Tech Stack: Python, TypeScript, LangGraph, LLM APIs, Multi-Agent system, Agentic RAG, AST, VS Code Extension API, Fine-tuning, Prompt engineering, Transformer Models, Pinecone.

BookFair Zone – Microservices Stall Reservation System

Built enterprise microservices platform for hall management, booking, real-time updatable map. JWT authentication, Kafka for event-driven communication. Implemented load balancer and API Gateway for traffic distribution.

Tech Stack: Java/Spring Boot, React, JPA/Hibernate, PostgreSQL, JWT, Kafka, Load Balancer, Api Gateway

Kafka Order Analytics Platform

Built real-time ordering system handling high-throughput message streaming with Apache Kafka, Avro serialization. Implemented fault-tolerant architecture, retry mechanisms and Dead Letter Queue. Live price aggregation in dashboard.

Tech Stack: Apache Kafka, Avro, Spring Boot, React, WebSocket, PostgreSQL

High-Security E-Voting System

Developed voting platform with end-to-end encryption and tamper-proof architecture. Implemented multi-factor authentication, JWT, token blacklisting, digital signature verification, and brute-force protection with rate limiting.

Tech Stack: Java, Spring Security, JWT, Role-based Auth, Multi Factor Auth, Email Verification, Cryptography

[Sinhala Sign Language Learning Management System](#)

Developed AI-powered LMS for hearing and speech impaired individuals, achieved 87% gesture recognition accuracy in real-time hand gestures using computer vision with MediaPipe pose estimation and deep learning models.

Tech Stack: Python, Computer Vision, OpenCV, MediaPipe, Deep Learning, TensorFlow, FastAPI, NLP

[HPC Optimization for LLM Parallel Prompt Processing](#)

Optimized LLM prompt processing using HPC techniques achieving 3.27x speedup. Implemented and benchmarked OpenMP, MPI, and hybrid parallelization strategies across Pure Python, Pure C, and C + Python processing prompts.

Tech Stack: C, Python, OpenMP, MPI, HPC Techniques, Performance Optimization, API Integration

[Multi-Tenant SaaS POS System](#) | [Frontend](#) | [Backend](#)

Developed multi-tenant point-of-sale system with database-level tenant isolation supporting unlimited business clients. Built role-based authentication system using Clerk and implemented modern responsive UI using shadcn ui.

Tech Stack: Nestjs, React, Next.js, Node.js, Express, MongoDB, Clerk Auth, shadcn/ui, Tailwind CSS, Docker

[Jenkins pipeline for a Spring Boot application](#)

Pipeline: Checkout, Test & Build, SonarQube Analysis, Quality Gate, Package, Docker Build, Deploy, and monitoring.

Tech Stack: Java 21, Spring Boot, PostgreSQL, Jenkins, Docker, SonarQube, Prometheus, Grafana, Gradle

EDUCATION

B.Sc. (Hons) Computer Engineering.

Faculty of Engineering, University of Ruhuna

GPA: 3.30

CERTIFICATES

[Java: Complete Java Core for Beginners with Exercises - 2025](#) - Andrill Partola ITER Academy, Udemy

[Spring Framework for Java Developers: Practical Guide](#) - Andrii Piatakha, Learn-IT University, Udemy

[AWS Essentials: A Complete Beginner's Guide](#) - Learnify IT, Udemy

[CI/CD with Jenkins and Docker](#) - Gourav J. Shah, School of Devops, Udemy

[Master in Software Architecture, Engineering and Development](#) - Arun Singhal B-Tech, MBA (IIM-B), Udemy

[Learn Spring GraphQL](#) - Ezekiel Eromosei, Udemy

[Programming Network Applications in Java- Cyber Quince](#), Udemy

[Next.js With Tailwind CSS-Build a Frontend Ecommerce Project](#)- Vapa Academy, Udemy

[Information Security Fundamentals](#) – Cyber Quince, Udemy

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, TypeScript, JavaScript
- **Backend:** Spring Boot, Spring Cloud, Node.js, NestJS, FastAPI
- **Frontend:** React, Next.js, Angular, HTML5, CSS3, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j, Pinecone
- **Software engineering:** Apache Kafka, RabbitMQ, Junit, Mockito, Unit Testing
- **DevOps & Cloud:** Docker, Kubernetes, Jenkins, AWS, CI/CD
- **Architecture:** Microservices System Design, RESTful APIs, gRPC, Event-Driven Systems, Server-Sent Events
- **AI/ML:** LangChain, LangGraph, RAG Systems, Multi-Agent Systems, Vector Databases, Fine-tuning

SOFT SKILLS

Communication & Presentation | Stress Management | Time Management | Team Collaboration | Leadership

INVOLVEMENT

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| • 3rd Year Field Representative – Computer Engineering | • Haxtreme 2.0 - 4th place |
| • IEEE INSL Competition – participation | • Former Senior Prefect – Mayurapada Central College |
| • Leadership Development program - Maliyadeva College | • Football Team Champion – Mayurapada Central |

REFERENCES

Dr.Kushan Sudheera - Senior Lecturer, Faculty of Engineering, University of Ruhuna.. +94719693164, kushan@eie.ruh.ac.lk

Dr. Rajitha Udawalpola - Senior Lecturer, Faculty of Engineering, University of Ruhuna. + 94718578608, rajitha@eie.ruh.ac.lk